Name: Roldan, Serge Ulver D.	**Date Performed: 26/02/2026**
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Instructor: Engr. Robin Valenzuela	**Semester and SY: Second Semester, S.Y. 2025 – 2026**

Activity 7: Managing Files and Creating Roles in Ansible

1. **Objectives:**

1.1 Manage files in remote servers

1.2 Implement roles in ansible

2. **Discussion**:

In this activity, we look at the concept of copying a file to a server. We are going to create a file into our git repository and use Ansible to grab that file and put it into a particular place so that we could do things like customize a default website, or maybe install a default configuration file. We will also implement roles to consolidate plays.

Task 1: Create a file and copy it to remote servers

1. Using the previous directory we created, create a directory, and named it "*files*." Create a file inside that directory and name it "*default_site.html*." Edit the file and put basic HTML syntax. Any content will do, as long as it will display text later. Save the file and exit.
2. Edit the *site.yml* file and just below the *web_servers* play, create a new file to copy the default html file for site:

```
- name: copy default html file for site

  tags: apache, apache2, httpd
  copy:
    src: default_site.html
    dest: /var/www/html/index.html
    owner: root
    group: root
    mode: 0644
```

3. Run the playbook *site.yml*. Describe the changes.

```
ask become pass

TASK [copying default html file] ***********************************************************
changed: [192.168.56.112]

PLAY RECAP *********************************************************************************
192.168.56.112             : ok=7    changed=3    unreachable=0    failed=0    skipped=0    resc
  ignored=0
192.168.56.113             : ok=5    changed=3    unreachable=0    failed=0    skipped=0    resc
  ignored=0
192.168.56.117             : ok=5    changed=2    unreachable=0    failed=0    skipped=0    resc
  ignored=0
```

4. Go to the remote servers (*web_servers*) listed in your inventory. Use cat command to check if the index.html is the same as the local repository file (*default_site.html*). Do both for Ubuntu and CentOS servers. On the CentOS server, go to the browser and type its IP address. Describe the output.

```
serge@node1: ~
serge@node1:~$ cat /var/www/html/index.html
<!DOCTYPE html>
<html>
<head>
    <title>Hello World Page</title>
</head>
<body>
    <h1>Hello World!</h1>
</body>
</html>serge@node1:~$
```

5. Sync your local repository with GitHub and describe the changes.

Task 2: Download a file and extract it to a remote server

1. Edit the site.yml. Just before the web_servers play, create a new play:

```
- hosts: workstations
  become: true
  tasks:

  - name: install unzip
    package:
      name: unzip

  - name: install terraform
    unarchive:
```

src: https://releases.hashicorp.com/terraform/0.12.28/terraform_0.12.28_linux_amd64.zip

dest: /usr/local/bin

remote_src: yes

mode: 0755

owner: root

group: root

2. Edit the inventory file and add workstations group. Add any Ubuntu remote server. Make sure to remember the IP address.
3. Run the playbook. Describe the output.

```
PLAY RECAP *********************************************************************************
192.168.56.112             : ok=0    changed=0    unreachable=1    failed=0    skipped=0    rescue
  ignored=0
192.168.56.113             : ok=0    changed=0    unreachable=1    failed=0    skipped=0    rescue
  ignored=0
192.168.56.117             : ok=0    changed=0    unreachable=1    failed=0    skipped=0    rescue
  ignored=0
192.168.56.119             : ok=4    changed=1    unreachable=0    failed=0    skipped=0    rescue
  ignored=0
```

4. On the Ubuntu remote workstation, type terraform to verify installation of terraform. Describe the output.

playbook.yml M inventory.ini M

roldan_sysad2_hoa6.1 > playbook.yml > { } 3 > [ ] tasks

```yaml
- hosts: file_server
  roles:
  - file_server

- hosts: workstations
  become: true
  tasks:
  - name: install unzip
    package:
      name: unzip
  - name: install terraform
    unarchive:
      src: https://releases.hashicorp.com/terraform/0.12.28/terraform_0.12.28_linu
      dest: /usr/local/bin
      remote_src: yes
      mode: 0755
      owner: root
      group: root
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS ANSIBLE bash - roldan_sysad2_hoa6.1

```
serge@ansible-control:~/ansible/roldan_sysad2_hoa6.1$ ansible-playbook -i inventory.ini playbook.y
ask-become-pass
fatal: [192.168.56.112]: FAILED! => {"changed": false, "msg": "Unable to start service apache2: Jo
 apache2.service failed because the control process exited with error code.\nSee \"systemctl statu
che2.service\" and \"journalctl -xeu apache2.service\" for details.\n"}

PLAY RECAP *********************************************************************************
192.168.56.112             : ok=8    changed=2    unreachable=0    failed=1    skipped=0    rescue
  ignored=0
192.168.56.113             : ok=0    changed=0    unreachable=1    failed=0    skipped=0    rescue
  ignored=0
192.168.56.117             : ok=0    changed=0    unreachable=1    failed=0    skipped=0    rescue
  ignored=0
```

Task 3: Create roles

1. Edit the site.yml. Configure roles as follows: (make sure to create a copy of the old site.yml file because you will be copying the specific plays for all groups)

```
---
- hosts: all
  become: true
  pre_tasks:

  - name: update repository index (CentOS)
    tags: always
    dnf:
      update_cache: yes
    changed_when: false
    when: ansible_distribution == "CentOS"
  - name: install updates (Ubuntu)
    tags: always
    apt:
      update_cache: yes
    changed_when: false
    when: ansible_distribution == "Ubuntu"

- hosts: all
  become: true
  roles:
    -  base

- hosts: workstations
  become: true
  roles:
    - workstations

- hosts: web_servers
  become: true
  roles:
    - web_servers

- hosts: db_servers
  become: true
  roles:
    - db_servers

- hosts: file_servers
  become: true
  roles:
    - file_servers
```

Save the file and exit.

2. Under the same directory, create a new directory and name it roles. Enter the roles directory and create new directories: base, web_servers, file_servers,

db_servers and workstations. For each directory, create a directory and name it tasks.

3. Go to tasks for all directory and create a file. Name it main.yml. In each of the tasks for all directories, copy and paste the code from the old site.yml file. Show all contents of main.yml files for all tasks.

playbook.yml M | main.yml .../base/... U | main.yml .../db_servers/... U | main.yml .../file_servers/... U

roldan_sysad2_hoa6.1 > roles > base > tasks > main.yml > { } 1 > abc name

Ansible Tasks Schema - Ansible tasks file (ansible.json)

```yaml
---
- name: update repository index (CentOS)
  tags: always
  dnf:
    update_cache: yes
  changed_when: false
  when: ansible_distribution == "CentOS"

- name: install updates on Ubuntu
  tags: always
  apt:
    update_cache: yes
  when: ansible_distribution == "Ubuntu"

- name: install unzip
  package:
    name: unzip
- name: install terraform
```

playbook.yml M | main.yml .../base/... U | main.yml .../db_servers/... U | main.yml .../file_servers/... U

roldan_sysad2_hoa6.1 > roles > db_servers > tasks > main.yml > { } 4 > { } copy > # mode

```yaml
- name: install terraform
  unarchive:
    mode: 0755
    owner: root
    group: root

- name: copying default html file
  tags: apache,apache2,httpd
  copy:
    src: default_site.html
    dest: /var/www/html/index.html
    owner: root
    group: root
    mode: 0644
```

Screenshot captured

You can paste the image from the clipboard.

main.yml .../file_servers/... U

roldan_sysadz_hoa0.1 > roles > file_servers > tasks > main.yml > [ ] 4 > [ ] copy > # mode

```yaml
  - name: install terraform
    unarchive:
      mode: 0755
      owner: root
      group: root

  - name: copying default html file
    tags: apache,apache2,httpd
    copy:
        src: default_site.html
        dest: /var/www/html/index.html
        owner: root
        group: root
        mode: 0644
```

Screenshot captured

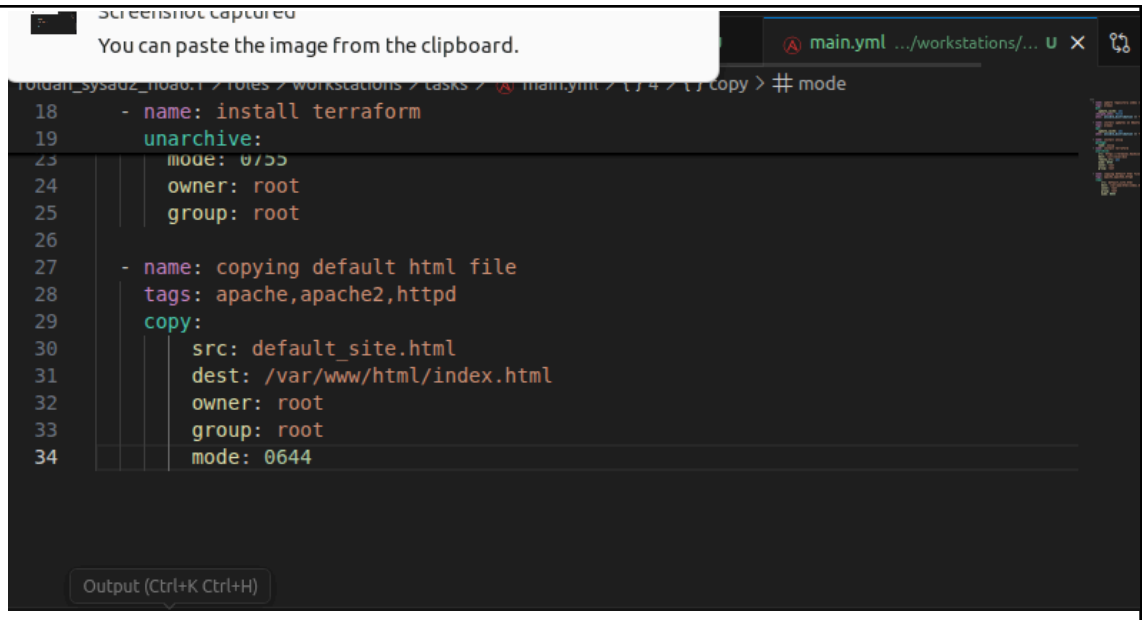
You can paste the image from the clipboard.

main.yml .../web_servers/... U

roldan_sysadz_hoa0.1 > roles > web_servers > tasks > main.yml > [ ] 4 > [ ] copy > # mode

```yaml
  - name: install terraform
    unarchive:
      mode: 0755
      owner: root
      group: root

  - name: copying default html file
    tags: apache,apache2,httpd
    copy:
        src: default_site.html
        dest: /var/www/html/index.html
        owner: root
        group: root
        mode: 0644
```



4. Run the site.yml playbook and describe the output.

playbook.yml M | main.yml .../base/... U | main.yml .../db_servers/... U | main.yml .../file_servers/... U

roldan_sysad2_hoa6.1 > playbook.yml > { } 1 > [ ] roles > abc 0

Ansible Playbook - Ansible playbook files (ansible.json)

```yaml
---

- hosts: all
  become: true
  pre_tasks:

  - name: update repository index (CentOS)
    tags: always
    dnf:
      update_cache: yes
    changed_when: false
    when: ansible_distribution == "CentOS"

  - name: install updates on Ubuntu
    tags: always
    apt:
      update_cache: yes
    when: ansible_distribution == "Ubuntu"
```

Terminal (Ctrl+`)

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS ANSIBLE bash - roldan_sysad2_hoa6.1

```
serge@ansible-control:~/ansible/roldan_sysad2_hoa6.1$ ansible-playbook -i inventory.ini playbook.yml -K
TASK [web_server : start apache2] ***********************************************************
fatal: [192.168.56.112]: FAILED! => {"changed": false, "msg": "Unable to start service apache2: Job for
 apache2.service failed because the control process exited with error code.\nSee \"systemctl status apa
che2.service\" and \"journalctl -xeu apache2.service\" for details.\n"}

PLAY RECAP *********************************************************************************************
192.168.56.112             : ok=14   changed=3    unreachable=0    failed=1    skipped=2    rescued=0
  ignored=0
192.168.56.113             : ok=11   changed=6    unreachable=0    failed=0    skipped=2    rescued=0
  ignored=0
192.168.56.117             : ok=11   changed=6    unreachable=0    failed=0    skipped=2    rescued=0
  ignored=0
```

Github Repo Link: https://github.com/Seijuro2004/roldan_sysad2_hoa6.1.git

Reflections:

Answer the following:

1. What is the importance of creating roles?

   Implementing roles improves structure, reusability, and scalability when it comes to automation. It turns monolithic playbooks into flexible, modular playbooks.

2. What is the importance of managing files?

   Proper file management ensures that systems stay consistent, secure, and reproducible while improving security, scalability, and simplifies maintenance.